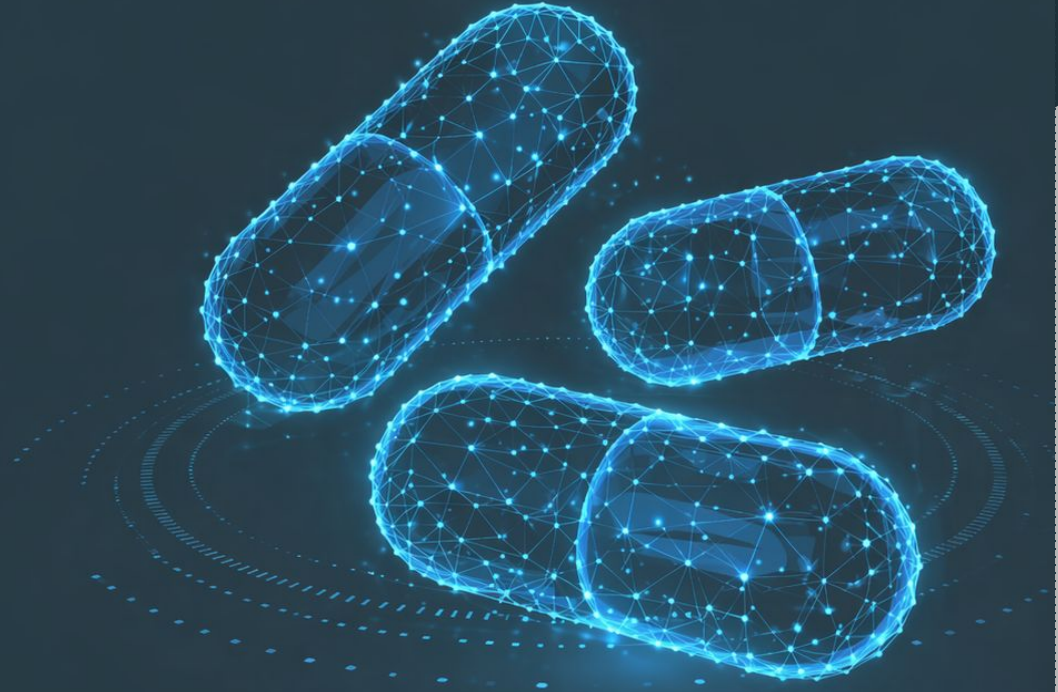


Drug related Side Effects Prediction Project



Objective

Develop a machine learning system that predicts **adverse drug reactions (ADRs)** based on known information about a drug.

Problem & Motivation

Adverse drug reactions are a major healthcare

- challenge because many drugs can produce multiple side effects, while traditional safety discovery is often slow and expensive.

Machine learning can help identify likely risks

- earlier, supporting safer prescribing, more efficient drug development, and lower healthcare burden.

This matters for:

- **Patients:** Improving safety and quality of life.
- **Doctors & Pharmacists:** Supporting safer decisions.
- **Researchers:** Identifying risks earlier.
- **Healthcare Systems:** Reducing complications and costs.

Current Approach

How ADRs are identified today

- Clinical trials
- Post marketing surveillance
- Drug safety databases (e.g., SIDER)
- Manual expert review

Common ML-based approaches

- Literature mining
- Statistical association methods
- Classification models (drug-side effect pairs)

Our Approach

(What's new and why it may succeed)

- Machine learning to ADR prediction using available drug information
- Data driven pattern recognition to detect potential side effect risks

- Improves efficiency compared with slower manual safety discovery approaches
- Supports earlier systematic screening of potential adverse reactions

Dataset

Raw Data

SIDER dataset: marketed drugs and known adverse drug reactions

Includes drug, indication, side effect

Uses MedDRA terminology for standardized labels



Cleaning

Merged drugs, indications, and side-effects (~9M records)

Removed missing, duplicate, and rare records

Final cleaned data retained ~2M records



Prepared Dataset

Final cleaned dataset used for modeling

Selected random 500,000-row subset for feasible training

Large raw data was cleaned and reduced into a feasible modeling subset

Challenges

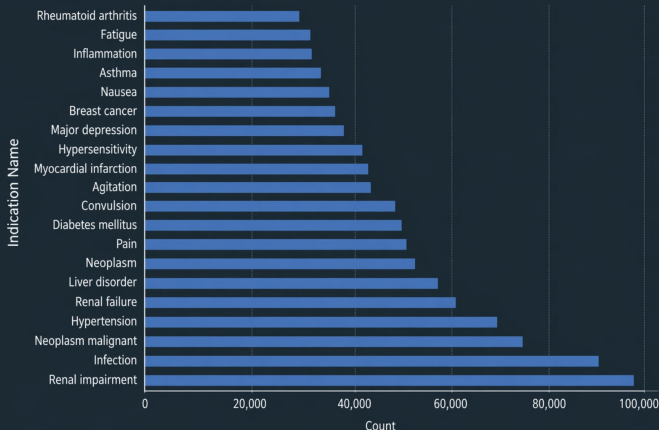
Large-scale data: full-data training required significant time and compute. Sampling was needed.

Multi-label nature: one drug can be associated with multiple side effects, which increases prediction complexity.

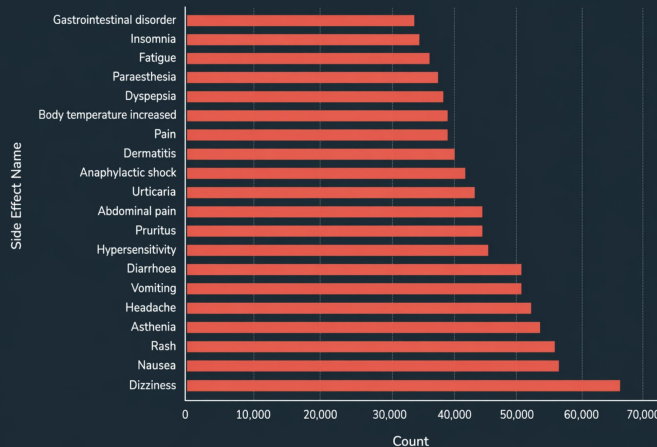
Complex relationships: the model must learn interactions between drugs, indications, and side effects.

Class imbalance: frequent labels dominate, while rare labels are harder to learn.

Top 20 Indication Names



Top 20 Side Effect Names



Model Selection

Approach	Key idea	Best F1	Decision
Baseline ML	Drug name + indication	~0.46	Too weak
Richer features	Add chemical features	~0.86	Strong improvement
Graph model (RGCN)	Model drug–indication–side effect relationships	~0.88	Final model ★

Baseline

- **Tested:** Logistic Regression, Small NN, XGBoost
- **Inputs:** drug name + clinical indication
- **Result:** performance was modest, so we explored stronger representations

Improvement directions

Richer features: added chemical features. Tested with Logistic Regression and small Neural Network

Advanced model: RGCN using drug–indication–side effect relationships

★ The RGCN showed **strong test performance**, supports **future scalability** such as drug–drug interaction modeling, and represents a **modern machine learning approach**, making it a strong final model choice.

RGCN Model Pipeline

1 Data Collection



- ADR records with drug indication, and side effect
- Merged data from multiple sources
- Standardized records for modeling

Raw Data

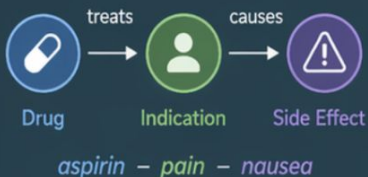
2 Pre-processing



- Removed missing values and duplicates
- Filtered to MedDRA preferred terms (PT)
- Cleaned and normalized text fields

Clean Data

3 Graph Construction



- Convert records into drug-indication-side effect triples
- Create node vocabularies and numeric IDs
- Build a heterogeneous graph
- Add relation types and reverse edges

Graph Structure

4 Model Training



- Train an RGCN on the relational graph
- Use positive and negative sampled triples
- Learn node embeddings through message passing

Trained Model

5 Prediction

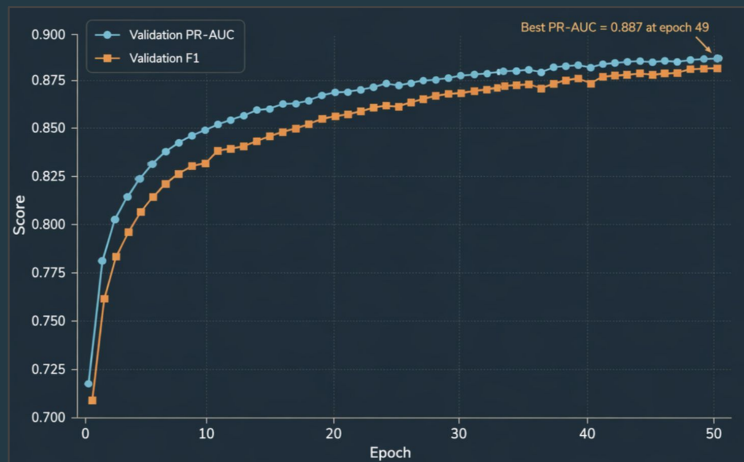


- Evaluated on validation and test sets
- Metrics: ROC-AUC, PR-AUC, F1, precision, recall
- Predict side effects for drug-indication pairs

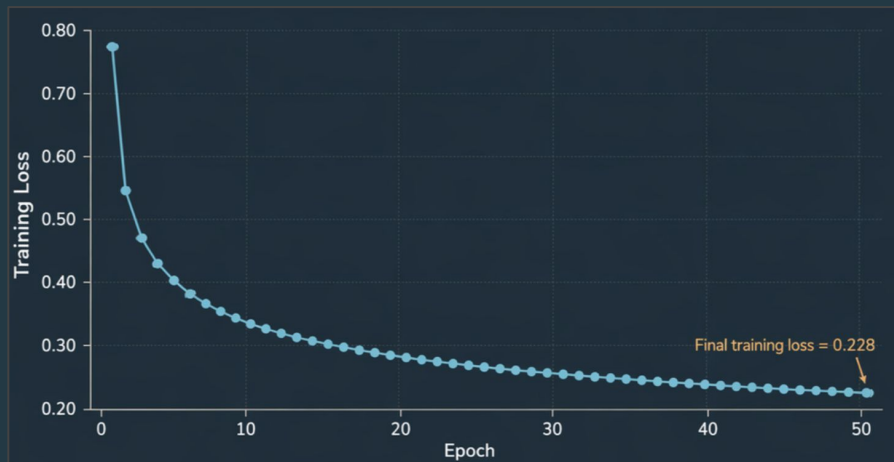
ADR Prediction

Training Progress

Validation PR-AUC and F1 across epochs



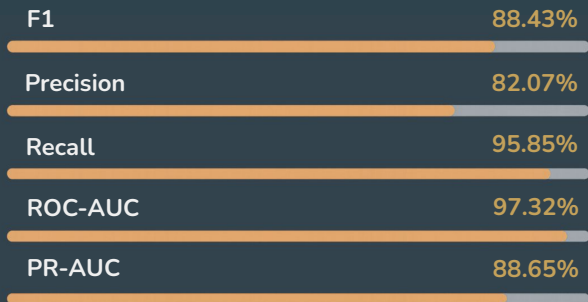
Training loss across epochs



- Stable optimization with steadily decreasing loss
- Validation PR-AUC improved to **0.887** at epoch 49
- Later epochs show smaller gains, indicating convergence

Evaluation & Results

RGCN Test metrics



Key Findings

- Strong predictive performance
- High recall, so fewer side effects are missed
- Good precision supports prediction reliability
- High ROC-AUC and PR-AUC show strong model discrimination

Evaluation Note

- Threshold tune on validation data
- Best checkpoint chosen based on validation PR-AUC
- Final metrics reported on a separate test set

Conclusion, Limitation & Future Work

Conclusion

- Graph-based learning was effective for ADR prediction.
- The final RGCN model was more scalable and flexible than simple baseline models.

Limitations

- Trained on a 500K-row subset, not the full dataset.
- Patient-specific factors were not included.
- Negative labels were created through sampling.

Future Work

- Train on larger datasets with more compute.
- Add drug-drug interactions and richer biomedical relations.
- Improve interpretability and probability calibration